**UNIVERSITY OF CAPE COAST**
**COLLEGE OF AGRICULTURAL AND NATURAL SCIENCES**
**SCHOOL OF PHYSICAL SCIENCES**
**DEPARTMENT OF PHYSICS**
**END OF FIRST SEMESTER EXAMINATION (2019/2020 ACADEMIC YEAR)**

**PHY 101N: GENERAL PHYSICS FOR NURSING** **DURATION: 2 HOURS**

***INSTRUCTIONS:*** *Answer all questions by choosing the most suitable answer from the options, lettered A-D. All other rough work should be done on the unused pages in the answer booklet.*

1. Which of the following physical quantities is measured in Joules?
   A. Time
   B. Energy
   C. Power
   D. Momentum

2. Each of the following are fundamental units except
   A. Kelvin
   B. Kilogram
   C. Watts
   D. Mole

3. Which of the following units is equivalent to the Newton?
   A. $ms^{-2}$
   B. $kgm^{-3}$
   C. $kgms^{-1}$
   D. $kgms^{-2}$

4. The BMI value recorded for a pregnant woman was determined to be 13.5 $kgm^{-2}$. The woman can be said to be
   A. Overweight
   B. Underweight
   C. Clinically obese
   D. Physically fit

5. The force needed to be applied to keep a body experiencing a net force of (8 N, 040°) in equilibrium is
   A. (0 N, 040°)
   B. (8 N, 040°)
   C. (8 N, 220°)
   D. (8 N, 050°)

6. A boy pulls his toy car with a force of 25 N inclined at 56° to a horizontal floor. The effective force pulling the toy along the floor is
   A. 41.0 N
   B. 22.5 N
   C. 12.1 N
   D. 10.9 N

7. The speed limit of a vehicle approaching a hospital is 25 $kmh^{-1}$. This is equivalent to
   A. 2.4 $ms^{-1}$
   B. 8.3 $ms^{-1}$
   C. 41.7 $ms^{-1}$
   D. 416.7 $ms^{-1}$

8. The following are all vector quantities except
   A. velocity
   B. Force
   C. volume
   D. displacement

9. The location of the centre of gravity of a body determines its
   A. density
   B. weight
   C. stability
   D. centre of mass

10. For the human ear, sound waves of frequencies greater than 20 kHz are said to be
   A. Audible
   B. Infrasonic
   C. Ultrasonic
   D. Supersonic

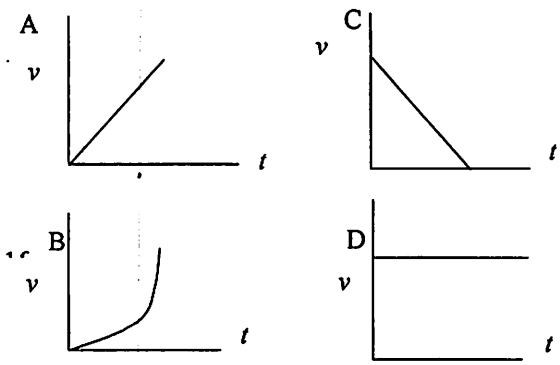
11. The opposition to the motion of a steel ball dropped in a tube filled with motor oil can be explained by the concept of
   A. friction
   B. viscosity
   C. elasticity
   D. surface tension

12. A 15 kg metal load is allowed to roll down a smooth plane inclined at 30° to the horizontal. Which of the following best represents its acceleration?
   A. 9.8 $ms^{-2}$
   B. 7.5 $ms^{-2}$
   C. 5.7 $ms^{-2}$
   D. 2.3 $ms^{-2}$

13. To every action, there an equal but opposite reaction. This is a statement of
    A. The law of floatation
    B. Newton's second law of motion
    C. Newton's third law of motion
    D. The law of conservation of momentum

14. Which of the following describes the closeness of a set of measured values to one another?
    A. Uncertainty
    B. Instrumental Error
    C. Precision
    D. Accuracy

15. A ball is dropped vertically downward to fall freely in air. Which of the following velocity-time graphs represents the ball's motion?




16. How far will a train travel if it moves at an average speed of 150 km/h in 40 minutes?
    A. 90 km
    B. 100 km
    C. 110 km
    D. 190 km

17. A stone is thrown vertically upwards with a velocity of 40 $ms^{-1}$. Calculate the total time taken to return to the thrower. (Take g = 10 $ms^{-2}$).
    A. 4 seconds
    B. 8 seconds
    C. 30 seconds
    D. 40 seconds

18. A load falls from a helicopter from and takes 24 seconds to arrive on the ground. How high is the helicopter about the ground? (Take g = 10 $ms^{-2}$).
    A. 360 m
    B. 240 m
    C. 180 m
    D. 120 m

19. The earth' s gravitational pull on objects

A. is the same at all places on the surface of the earth
B. is greater at the equator of the earth than at the poles
C. decreases with altitude
D. increases with altitude

20. Which of the following components of the electromagnetic spectrum have been correctly arranged in order of increasing frequency?
A. Infrared, Microwaves, Ultraviolet, Radio waves
B. X-rays, Infrared, Ultraviolet, Visible,
C. Radio waves, Microwaves, X-rays, Gamma
D. Ultraviolet, Infrared, Radio waves, Gamma

21. Which of the following statements is/are true of the thermal energy supplied to a substance during a change of state? It is used to
I. increase the kinetic energy of the molecules
II. break the bonds between molecules
III. increase the temperature of the substance
A. I only
B. II only
C. I and III only
D. I, II and III

22. Sound is a mechanical wave because
A. it is able to cause vibrations
B. it requires material medium for propagation
C. its velocity is less than that of light
D. its amplitude increases with distance travelled

23. The change in volume of a metal per unit volume per unit temperature change defines
A. coefficient of linear expansion
B. coefficient of superficial expansion
C. coefficient of cubical expansion
D. temperature coefficient of resistance

24. Calculate the quantity of heat required to raise the temperature of 400 g of a piece of aluminium from 25 °C to 50 °C. (Specific heat capacity = 920 $Jkg^{-1}K^{-1}$).
A. 4.6 kJ
B. 9.2 kJ
C. 13.8 kJ
D. 18.4 kJ

25. The boiling point of a thermometric liquid is 172 °F. What is the equivalent of this temperature in degrees Celsius?
    A. 204
    B. 140
    C. 78
    D. -8

26. The rate of change of momentum is proportional to the applied force and takes place in the direction of the force. This statement is about
    A. Newton's second law of motion
    B. The principle of conservation of momentum
    C. The principle of moments
    D. Newton's first law of motion

27. Which of the following instruments is not required by a nurse in taking the vital statistics of a patient?
    A. Barometer
    B. Thermometer
    C. Weighing balance
    D. Sphygmomanometer

28. The resistance of a conducting wire depends increases with temperature. This is the thermometric property used in the construction of a
    A. Clinical thermometer
    B. Platinum Resistance thermometer
    C. Thermocouple
    D. Pyrometer

29. Which of the following is NOT a correct representation of Newton's equations of motion?
    A. $s = ut + \frac{1}{2}\ at^2$
    B. $s = \frac{v+u}{2}t$
    C. $v = u + at$
    D. $v^2 = a^2 - 2us$

30. Any pressure applied to an enclosed fluid is transmitted equally in all directions throughout the fluid. This is a statement of the
    A. Archimedes Principle
    B. Bernoulli's law
    C. Reynold's law
    D. Pascal's law

31. This listening device used by health professionals to check the pulse rate of patients.
A. Barometer
B. Centrifuge
C. Stethoscope
D. Sphygmomanometer

32. The systolic pressure recorded for a hypertensive patient is about
A. 140 mmHg
B. 120 mmHg
C. 90 mmHg
D. 60 mmHg

33. Which type of wave does not require a material medium for propagation?
A. Electromagnetic
B. Longitudinal
C. Mechanical
D. Transverse

34. The area under a velocity-time graph of a body represents
A. the maximum speed attained
B. the uniform acceleration
C. the total distance covered
D. the work done by the body

35. At 4° C, freezing water attains its
A. minimum density
B. minimum volume
C. freezing point
D. boiling point

36. The following are all characteristics of a laser beam except
A. monochromaticity
B. non-divergence
C. incoherence
D. high intensity

37. A body of mass 500 g is dropped from a height of 40 m. With what energy does it strike the ground? [Take g = 10 $ms^{-2}$]
A. $10^{-2}$ J
B. $10^{-1}$ J
C. $10^{2}$ J
D. $10^{3}$ J

38. Calculate the power of a pump which lifts 2000 kg of water through a vertical distance of 5 m in 10 s. [Take g = 10 $ms^{-2}$]
    A. 50 kW
    B. 100 kW
    C. 5000 kW
    D. 10000 kW

39. Excessive exposure to which of the following radiations can cause skin cancer?
    A. Ultraviolet
    B. Infrared
    C. Gamma
    D. X-rays

40. Steam burns are more severe than hot water burns because
    A. Vapour has higher temperature than boiling water
    B. Vapour molecules are more than liquid molecules
    C. The heat content in boiling water is less than that of steam
    D. The heat capacity of steam is less than that of liquid water

41. This radiation is widely used in sensing devices and night vision cameras.
    A. Microwaves
    B. Radio waves
    C. Infrared waves
    D. X-rays

42. When an applied force causes a body to a move at an obtuse inclination to the force, the work done is said to be
    A. conserved
    B. zero
    C. positive
    D. negative

43. A particle of mass 1.6 $x10^{-27}$kg, emerging from a nuclear reactor travels a distance of 4.5m across a research laboratory in 180ns.What is the kinetic energy of the neutron?
    A. 5.0 x $10^{-13}$ J
    B. 7.2 x $10^{-18}$ J
    C. 5.0 x $10^{-18}$ J
    D. 4.0 x $10^{-13}$ J

44. The opposing force that exists between particles during a fluid flow is referred to as
    A. upthrust
    B. laminar flow
    C. Brownian motion
    D. viscosity

45. Wearing of sun glasses is meant to protect the eye from the harmful effects of which of the following radiation?
    A. Visible light
    B. X-rays
    C. Ultraviolet rays
    D. Radio waves

46. The quantity of heat required to change the temperature of a unit mass of a substance through 1 °C is known as
    A. latent heat
    B. heat capacity
    C. specific latent heat
    D. specific heat capacity

47. This type of thermometer may be recommended for use by doctors in an Ebola-infected institution?
    A. Pyrometer
    B. Thermocouple
    C. Platinum Resistance
    D. Constant-Volume Gas

48. A man standing 100.8 m away from a cliff fires a gun and hears an echo 0.6 s later. Find the velocity of sound in air.
    A. 122 $ms^{-1}$
    B. 168 $ms^{-1}$
    C. 336 $ms^{-1}$
    D. 340 $ms^{-1}$

49. The rate of change of momentum is proportional to the applied force and takes place in the direction of the force. This statement is about
    A. the principle of conservation of energy
    B. Newton's second law of motion
    C. the principle of conservation of momentum
    D. Newton's first law of motion

50. A train recedes from a stationary observer at 30 $ms^{-1}$ while blowing a horn at 512 Hz. What is the apparent frequency heard by the observer, if the velocity of sound in air is 330 $ms^{-1}$?
    A. 563 Hz
    B. 482 Hz
    C. 465 Hz
    D. 469 Hz

51. Laser light is used in surgery as a cutting tool because of its
    A. Non-divergence over long distances
    B. Brightness or high intensity
    C. Ability to be focused to a microscopic spot
    D. Monochromaticity

52. An endoscope is made from glass material of refractive index 1.62. What is the critical angle of the material?
    A. 38°
    B. 45°
    C. 52°
    D. 56°

53. Which of the following animals is likely to perceive sound at ultrasonic frequencies?
A. Elephant
B. Bat
C. Cow
D. Dog

54. Which of the following pairs of quantities are dimensionally equivalent?
    A. Heat and Temperature
    B. Mass and Tension
    C. Energy and Power
    D. Weight and Force

55. Blood-thinning drugs are administered to clients to reduce the
    A. red blood cell count
    B. white blood cell count
    C. viscosity
    D. pulse rate

56. An athlete will struggle to inhale at a location where the barometer reading is
    A. 742 mmHg
    B. 760 mmHg
    C. 770 mmHg
    D. 780 mmHg

57. The earth's gravitational pull on objects

    A. is the same at all places on the surface of the earth
    B. is greater at the equator of the earth than at the poles
    C. decreases with altitude
    D. increases with altitude

58. Which of the following thermometers is best suited for extremely high temperature environments in industry?

A. Clinical thermometer
B. Platinum-resistance thermometer
C. Thermocouple
D. Six's maximum and minimum thermometer

59. The quantity of heat generated by an electric current is determined by Joule's heating represented as

A. $P = V I$
B. $V = I R$
C. $H = I^2 R t$
D. $Q = m c \Theta$

60. The standard unit for commercialising electrical energy is the

A. Kelvin
B. kilowatt hour
C. kilojoule per second
D. ohm-meter